

Demystifying THERMOPILE SENSORS



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Thermopile temperature sensors offer the advantage of non-contact temperature measurement, which has made them more popular than traditional contact-type temperature sensors. These sensors use infrared (IR) radiation for heat transfer, and are commonly used in portable devices such as infrared thermometers. Another attractive application for thermopile sensors is temperature monitoring of moving objects, where contact-based temperature sensors have serious drawbacks. Here is an overview of thermopile infrared temperature sensors to help you take full advantage of this technology.

Sensor insight

A thermopile infrared sensor has a number of thermocouples connected in series that have their 'hot' junctions attached to a thin infrared absorber, usually on a micro-machined membrane on a silicon chip. The exchange of infrared radiation between the absorber and the object in front of it makes the temperature of the absorber rise or fall, depending on the temperature difference between it and the object.

To properly measure the object's temperature, the object must completely fill the sensor's field of view (FOV).

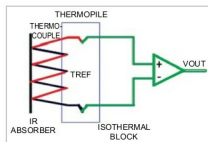


Fig. 1: Structure of a thermopile

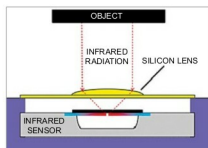


Fig. 2: Silicon lens/filter execution

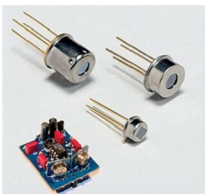


Fig. 3: Thermopile sensors/modules

This ensures that the infrared radiation affecting the thermopile is coming only from the object of interest and not its background. Besides, use of a filter and lens greatly enhances the performance of infrared sensors.

Plain silicon is completely opaque to visible light, but transparent to wavelengths longer than $2\mu\text{m}$ where most spectral emission occurs for temperatures below 500K (200°C or 450°F). Therefore silicon material can be used for filtering the visible and ultraviolet (UV) spectrum from affecting the sensor. In order to increase the sensor's sensitivity (or the distance from which the sensor can measure a fixed-size object), micro-machined silicon lens is widely used to concentrate more infrared radiation onto the sensor or limit its field of vision.

Sensor playground

Nowadays, thermopile infrared (IR) temperature sensors are available with various lenses/filters, allowing use in real-world applications ranging from industrial pyrometers to lifecare devices. Depending on the sensor, the output signal can be the standard resistance output or analogue/digital signal output.

Various sensor packages (useful for hobbyists and makers alike), including hermetically sealed housings, stainless-steel packaged housings and pre-wired

modules, are now available at affordable prices. All of these thermopiles are designed to measure temperature from a distance by detecting the object's infrared energy. The thermopile sensing element, composed of small thermocouples on a silicon chip, absorbs the energy and produces an output signal. A reference sensor is also included in the package as a reference for compensation.

TPS334 thermopile detector.

TPS334 thermopile detector, from Excellitas, is a standard sensor that